

关于AI和机器人的想象，很多都是不切实际的

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在每个产业转移，全[产业链](#)，工业化等问题下面，都充斥着认为自动化会让工厂里的体力劳动变得无足轻重的观点。这种观点可以用来论证1) 中国的产业不用转移 2) 欠发达国家的廉价劳动力优势无法发挥 3) 等等

在Chat GPT+发布之后，AI的能力让人们产生了很多想象。很多人都认为，AI会让绝大多数白领失业。同样的，[自动驾驶+](#)和机器人又会让蓝领失业。人类劳动力成为多余的，多数人都没有工作了，最终要靠基本收入之类的制度来解决[分配问题+](#)。

有人甚至相信马斯克和[奥特曼+](#)之流的说法，认为AGI即[通用人工智能+](#)即将诞生。而把AGI作为大脑加到机器人上就成了[通用机器人+](#)。加上各种机器人公司搞了各种花里胡哨的展示，让很多人以为，所谓“未来以来”。

今天我们先不探讨，探讨通用机器人出现后，资本主义文明的演进问题。这个问题很大，我们后面在讨论。这篇文章主要想探讨一下当前这轮AI和机器人技术发展的局限性。

LLM

因为我工作会用到[大模型+](#)来做一些工作的自动化，在实际使用过程中，发现大模型的能力上存在如下特征：

- 1) 在做大量Token输入，少量token输出的工作表现特别好：如[API+](#)选择，参数提取，基于添加了大量背景提示词的[任务规划+](#)
- 2) 大量token输入，大量token输出，但是无创作成分：如翻译
- 3) 少量token输入，大量token输出，无逻辑要求：如文学艺术创作类

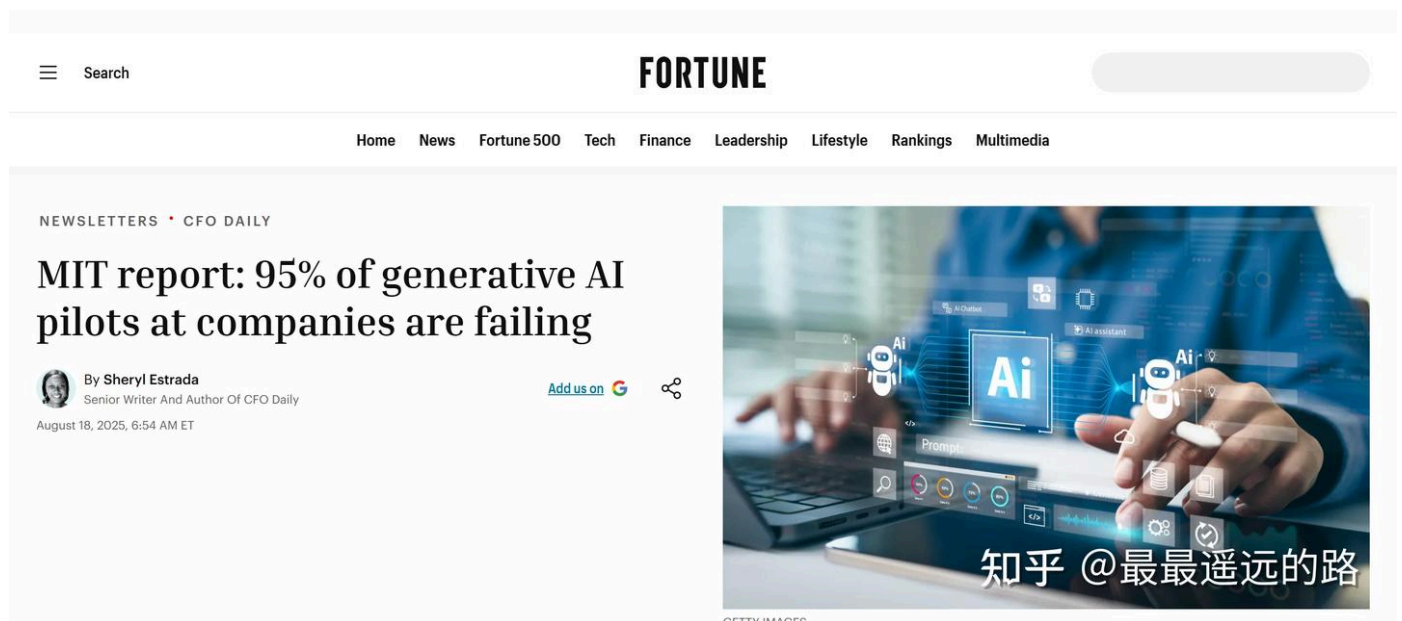
这些工作已经可以在很大程度上代替人类劳动。但是，基于LLM的这些能力，融入到工作流程中，本身需要大量定制，无论是定制提示词，还是做[模型微调+](#)，都是如此。而且，在今年初之后的模型，似乎没有太大的进步。无论经过多少微调，优化，[工程方法+](#)，实际商用的场景能有90%的正确率就很好了。

比如，我们之前考虑过直接根据需求描述输出代码，而且还是选取的一个针对coding做过专门训练的模型。但实际上，如果你给的背景信息不足，写出来的代码总是缺这缺那，只能不断地交流才能得到合格的代码。而且这里的合格代码只是说基本用例跑通，不代表没有BUG+。用过的人都知道，AI写出来的代码BUG一堆。基于这种LLM Brain做出来的产品，要把人代替，那是不可能的。

LLM能代替什么呢？各种客服，流程类的白领。因为很多工作其实就是用自然语言从客户或上下游提取参数，然后填入某个IT系统，这类工作用LLM代替那确实是容易的。但也仅限于此了，想让大模型把大部分白领都搞失业是不可能的。

最近用来证明AI 泡沫+的几份报告也指出了这个问题。

比如MIT+的报告说明，95%的企业用生成式AI+都是失败的，也就是说，虽然很多企业都在跟风搞AI，但实际上用不起来。其他类似的报告有好几个，结论都大同小异。



那有人说，大模型的能力会不会继续提升呢？继续堆GPU+，增加参数量，训练出越来越大的模型。实际上，大模型靠scaling law 大力飞砖的路线已经走到头了。

给大家推荐一篇谷歌工程师+写的文章：

Fundamental Limits of Large Language Model Scaling: Analysis of Mathematical Impossibility Theorems and Resource Constraints

<https://medium.com/hackrlife/fundamental-limits-of-large-language-model-scaling-analysis-of-mathematical-impossibility-theorems-53ece210e23e>

文章里收集了最近出来的25篇论文，从幻觉控制的不可可能性，数据枯竭，合成数据的局限性，Performance Plateaus and Diminishing Returns (scaling 性能瓶颈+期和收益减少)，对齐的局限性等等方面论证，基于当前架构的LLM AI+就是这么个水准了，要靠scaling law继续提升，很难了。

比如如下论文就谈到，参数量和收益的不成正比。模型规模和性能提升之间是对数关系，模型的规模的每次翻倍带来的效益越来越低。

6.1 Emerging Patterns in Scaling Limitations

Our analysis has revealed several key patterns that characterize the current state of LLM scaling:

- The relationship between model size and performance improvements follows a logarithmic curve, with each doubling of parameters yielding progressively smaller gains. This pattern is particularly evident in benchmark results where GPT-4's 76.60% accuracy on math tasks and Claude 3.5 Sonnet's 90.20% accuracy in code evaluations (Vellum Research Team, 2024) represent significant but increasingly difficult-to-surpass achievements.
- The resource requirements for crossing higher emergence thresholds (θ_c) grow exponentially, while the corresponding performance improvements grow logarithmically. This misalignment between cost and benefit suggests that continued scaling alone cannot be the primary path forward.
- Data quality and diversity have emerged as potentially more critical constraints than raw computational power. The variance term $V(P, D)$ in our bias-variance decomposition increasingly dominates performance limitations, indicating that data innovation may be more crucial than model size scaling.

Has LLM Reached the Scaling Ceiling Yet? Unified Insights into LLM Regularities and Constraints

<https://arxiv.org/pdf/2412.16443>

如下文章，证明了幻觉不可避免，想让LLM的幻觉消失是不可能的。这不是LLM的架构，而是信息聚合+造成的，因而是不可避免的。

9 Summary

We have established that no large language model can simultaneously achieve perfect hallucination (or imagination) control. We showed that four essential properties—truthful knowledge representation, semantic information conservation, complete revelation of relevant knowledge, and optimal response generation—are mutually incompatible. This impossibility emerges from the mathematical structure of information aggregation itself, not from limitations in data, compute, or architecture.

We introduced a rigorous mathematical framework and grounded in it the general analysis of inference processes and their LLM implementations. At the heart of our mathematical framework lies the semantic information measure μ_C and the emergence operator $\mathcal{E}_C$. The earlier one is a context-dependent metric capturing how knowledge reduces uncertainty within computational bounds. The later one formalizes how reasoning makes latent knowledge explicit rather than creating it anew. Finally, we model LLM inference as an auction of ideas—a idealized marketplace where neural components compete by bidding with their partial knowledge, each trying to influence the final response.

The impossibility manifests through three complementary proofs, each illuminating different aspects of the fundamental constraint. Through game theory, we apply the Green-Laffont theorem to show the impossibility when knowledge components are independent. Through probability theory, we leverage Savage's proper scoring rules to extend the result to general trained or fine-tuned probability distributions. And through direct analysis of transformer architectures, we demonstrate how log-sum-exp convexity in attention mechanisms creates measurable violations of information conservation. The last proof exploits Jensen inequality for convex mappings to define the precise mathematical quantity that measures how aggregation of ideas (auction bids) produces overconfident responses that the constituent components cannot justify with revealed knowledge.

These mathematical insights we have made yield some interesting philosophical implications. Hallucination and creativity (two sides of the same coin we call imagination) emerge as mathematically identical phenomena, distinguished only by our normative judgments. The impossibility result explores the outer limits of reason, identified by Gödel's incompleteness in mathematics, Heisenberg's uncertainty in physics, and Arrow's impossibility in social choice. Also, our framework seems to provide mathematical foundations and experimental scaling for models of artificial consciousness. That may open interesting avenues for future research.

On the Fundamental Impossibility of Hallucination Control in Large Language

Models <https://arxiv.org/pdf/2506.06382>

其他的论文，有兴趣的朋友可以自己去读一读。

很巧的是，GPT的作者Ilya Sutskever最近也在说，Scaling的时代结束了，现在是回答实验室继续研究新架构的时候了。换句话说，本次AI商业落地爆发期已经结束，又要回到实验室，等到下一次技术突破，再开启新一轮的商业爆发期。他这么说可能有自己的一些动机，但是行业里早有人这么讲了。主要是他是Scaling Law的提出人，所以由他来宣判死刑才有权威性。

Ilya Sutskever Declares the Scaling Era Dead. His \$3 Billion Bet Says Research Will Win.

Ilya Sutskever helped prove that scale works in AI. Now he says it doesn't—and his \$3 billion company is betting the future belongs to researchers with ideas, not labs with the biggest GPU clusters. The industry isn't ready for this argument.



MARCUS SCHULER

NOVEMBER 25 2025 • 10:20 AM (UTC-08:00 • PST) • 10 MIN READ

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机器人

当前关于机器人最火的一个说法是，人形机器人能够适应于围绕着人类工人建立的一整套基础设施，可以平滑地替代人类劳动者。一旦人形机器人达成自己的“chatgpt时刻”，那么全球好几十亿体力劳动者都可以被替代，这将是一个巨大的市场。

我本人不是搞机器人的，没有相关经验。但是我一直对最近鼓吹机器人让流水线工人变得多余的说法持怀疑态度。

Robot factories could threaten jobs of millions of garment workers



📷 Workers on a production line in Ho Chi Minh City, Vietnam. Nine million people, mostly young women, are dependent on garment and footwear jobs in south-east Asia. Photograph: Richard Vogel/AP

As Adidas announces a new robot shoe factory, an ILO report says up to 90% of workers in south-east Asia could face unemployment due to automation

The jobs of nearly 90% of garment and footwear workers in Cambodia and Vietnam are at risk from automated assembly lines - or “sewbots” - according to a new [report](#) from the International Labour Organisation (ILO).

There are 9 million people, mostly young women, dependent upon jobs in textiles, garments, and footwear within the Association of Southeast Asian Nations (Asean) economic area, which includes [Cambodia](#), Indonesia, Thailand and Malaysia. These are the workers the ILO identifies as most

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但9年过去了，[孟加拉](#)和越南的服装厂工人被代替了吗？没有。光孟加拉都有500万[服装业](#)工人。

Over 50.17 lakh people working in garment sector in Bangladesh: state minister

Staff Correspondent 23 June, 2024, 21:39 [Best clothing retailers](#)

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There are over 50.17 lakh workers working in garment factories in the country and 55.57 per cent of them are women, state minister for labour and employment Nazrul Islam Chowdhury told Jatiya Sangsad on Sunday.

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最近浏览到一篇MIT机器人专业教授Rodney Brooks+的文章。他是澳大利亚科学院+院士，MIT计算机科学与人工智能实验室+主任，iRobot的创建者之一，在机器人领域+学术界和工业界工作了几十年，绝对是内行了。

他在Why Today' s Humanoids Won' t Learn Dexterity 这篇文章里谈到了两个重要的原因：

1，当前的人型机器人+学习都是基于视觉的，但是人手的灵活性不仅来自于视觉，更来自于触觉。没有触觉数据+训练的机器人系统，是很难做到“手”的灵活性的。但是这条路径的难点在于，触觉的训练数据相对于视觉很难获取。目前的实验室研究+都很初步，还谈不上真正的商用。

2，当前的人型机器人能力有限，不可能一上来就代替人，所以需要与人一起工作。但是人形机器人+和人在同一个工作空间里工作，靠的很近，其实是非常危险的。他的建议是不要离完全尺寸的

人形机器人太近...

My advice to people is to not come closer than 3 meters to a full size walking robot. And the walking robot companies know this too. Even in their videos you will not see people close to a locomoting humanoid robot unless there is a big table between them, and even then the humanoids only shuffle around a little bit.

按照他[技术预测](#)贴Predictions Scorecard, 2025 January 01 中的估计:

1) 2030年前, 不可能出现灵活的机器手, 他希望是在2040年前能搞出来。

Dexterous robot hands generally available.	NET 2030 BY 2040 (I hope!)	Despite some impressive lab demonstrations we have not actually seen any improvement in widely deployed robotic hands or end effectors in the last 40 years.
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2) 达到猫狗级别的机器人, 在2048年前不会出现。

A robot that seems as intelligent, as attentive, and as faithful, as a dog.	NET 2048	This is so much harder than most people imagine it to be--many think we are already there; I say we are not at all there.
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3) 能像一个6岁小孩那样理解人类的机器人, 在他有生之年(2050年前)看不到。

A robot that has NIML
any real idea
about its own
existence, or the
existence of
humans in the
way that a six
year old
understands
humans.

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这个预测会根据产业界的现状不断调整，但是可以明确的是，当前已经着眼于商业化的这些人形机器人，压根不具备替代人的实力，能否真的进入人类工作场所协助人类工作，都要打一个大大的问号。

自动驾驶

自动驾驶算是AI应用里最成熟的一种了，在很多眼里，似乎过几年司机这个工作就得消失了。但行业内的专家其实不这么看。

Rodney Brooks对自动驾驶的评估是不可能到2045年前全部商用。现在所谓商用的[自动驾驶汽车+](#)，要么里面做了个安全员，要么是后台有人操控，不是真正的自动驾驶。

The majority of US NET 2045
cities have the
majority of their
downtown under
such rules.

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自动驾驶的Hype已经出现好几轮了，目前是最新的一轮。

Some of the AI Hype Cycles that I Have Seen

- Pre-me:
 - Neural networks (vol 1), 1943, McCulloch & Pitts
 - Turing, 1950 computers fast enough for AGI
 - Neural networks (vol 2) 1950s, Minsky & Rosenblatt
 - McCarthy, 1956 computers fast enough for AGI
- Michie, 1959, reinforcement learning (1)
- Minsky, Michie, Nilsson, 1960: search
- It is all first order logic & resolution theorem proving
- Neural networks (vol 3) 1968 Minsky again
- MacHack (search, chess 1)
- Fuzzy logic, STRIPS, knowledge-based systems (revolution in medicine 1)
- Neural networks (vol 4) back propagation
- Primal sketch, 1979, Marr
- Self driving cars 1, 1987, Dickmanns
- 1980s reinforcement learning (2, & 3)
- SOAR, qualitative reasoning, support vector machines
- Self driving cars 2, 1997, Kanade et al
- Deep Blue (search, chess 2)
- Self driving cars 3, 2007, Thrun
- Bayesian inference
- Watson (Jeopardy & revolution in medicine 2)
- Deep learning, neural networks (vol 5)
- Self driving cars 4, 2010+, commercial
- Alpha Go, reinforcement learning (4)
- Generative AI, neural networks (vol 6)

各种车企吹完全自动驾驶，已经吹了很多年了，就跟中国车企吹自己能到L3，其实只能在XX场景下达成L3，一堆限定词之后，等于不是L3。

FORECASTS: http://www.driverless-future.com/?page_id=384 March 27, 2017

NVIDIA to introduce level-4 enabling system by 2018 (2017)
NuTonomy to provide self-driving taxi services in Singapore by 2018, expand to 10 cities around world by 2020 (2016)
Delphi and MobilEye to provide off-the-shelf self-driving system by 2019 (2016)
Ford CEO announces fully autonomous vehicles for mobility services by 2021 (2016) ←
Volkswagen expects first self driving cars on the market by 2019 (2016)
GM: Autonomous cars could be deployed by 2020 or sooner (2016) ←
BMW to launch autonomous iNext in 2021 (2016) ←
Ford's head of product development: autonomous vehicle on the market by 2020 (2016) ←
Baidu's Chief Scientist expects large number of self-driving cars on the road by 2019 (2016)
First autonomous Toyota to be available in 2020 (2015) ←
Elon Musk now expects first fully autonomous Tesla by 2018, approved by 2021 (2015)
US Sec Trans: Driverless cars will be in use all over the world by 2025 (2015)
Uber fleet to be driverless by 2030 (2015) ←
Ford CEO expects fully autonomous cars by 2020 (2015) ←
Next generation Audi A8 capable of fully autonomous driving in 2017 (2014)
Jaguar and Land-Rover to provide fully autonomous cars by 2024 says Director of Research and Technology (2014)
Fully autonomous vehicles could be ready by 2025, predicts Daimler chairman (2014) ←
Nissan to provide fully autonomous vehicles by 2020 (2013) ←
Truly autonomous cars to populate roads by 2028-2032 estimates insurance think tank executive (2012)
Continental to make fully autonomous driving a reality by 2025 (2012)

结语

在Scaling Law触到天花板的情况下，其他各种AI应用的限度其实也就确定下来了。无论是机器人，还是自动驾驶，要实现人类级别的灵活性，除了感知，还得有推理能力。推理要么是基于规则的，或者是靠神经网络。基于规则总有边边角角的场景考虑不全，基于神经网络+的如何解决不准确的情况。在白领的数字化工作领域里，可能还有得试。在体力劳动领域，因为涉及到人身安全问题，如何推广本身就是难题。但是不推广，如何获取数据用于进一步的训练？蛋生鸡，鸡生蛋，谁先谁后？

AI，机器人，自动驾驶等技术发展对于人类按道理是好事，但是这些技术的利用如果只是为了资本的目的，却可能给人类带来灾难。幸运的是，技术发展没有马斯克等人吹的那么厉害，人类还有机会，在更好的社会制度下，发展和应用AI技术。

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